DannyNet: Reproducing and Improving CheXNet for Multi-Label Thoracic Disease Classification on NIH ChestX-ray14 Using DenseNet-121 and Focal Loss

Abstract

Automated interpretation of chest radiographs remains one of the most clinically impactful applications of deep learning in medical imaging. This paper presents **DannyNet**, a faithful PyTorch reproduction and systematic improvement of CheXNet (Rajpurkar et al., 2017) applied to the NIH ChestX-ray14 dataset, one of the largest publicly available chest X-ray repositories comprising 112,120 frontal-view radiographs spanning 14 distinct thoracic disease classes. Our approach employs a DenseNet-121 backbone pre-trained on ImageNet, fully fine-tuned end-to-end with a multi-label Focal Loss objective ($\gamma=2$, $\alpha=1$) to address the severe class imbalance inherent in medical imaging datasets. We adopt patient-level train/validation/test splits (70/10/20) to eliminate data leakage and employ per-class decision threshold optimization on the validation set to maximize per-disease F1 scores. Training utilized the AdamW optimizer (lr=5×10^{-5}, weight_decay=1×10^{-4}) with a ReduceLROnPlateau scheduler over 20 epochs on dual Kaggle T4 GPUs. DannyNet achieves a mean test AUC of 0.831 across all 14 disease classes, with peak individual AUCs of 0.914 (Emphysema), 0.909 (Hernia), and 0.906 (Cardiomegaly). A per-class F1 of 0.312 is achieved following threshold tuning. Grad-CAM heatmap visualizations confirm that the model attends clinically meaningful anatomical regions, particularly for well-localized pathologies such as Cardiomegaly. Comparative analysis with the original CheXNet demonstrates improvement in 4 of 14 disease categories. These results demonstrate the viability and reproducibility of deep learning pipelines for multi-label thoracic disease classification and lay a foundation for future improvements using domain-adapted pretraining and advanced architecture.

Keywords: Chest X-ray classification, NIH ChestX-ray14, DenseNet-121, multi-label classification, Focal Loss, CheXNet, Deep learning, medical image analysis, Grad-CAM, Class imbalance, AdamW, Transfer learning, Thoracic disease detection, AUC-ROC

1. Introduction

Chest radiography (X-ray) is among the most frequently performed medical imaging procedures worldwide, with over two billion chest X-rays conducted annually. Despite its widespread use and clinical utility, accurate interpretation of chest radiographs demands highly specialized expertise in radiology, a resource that remains critically scarce in many regions across the globe. The global shortage of radiologists, compounded by increasing imaging volumes, underscores the urgent need for reliable, scalable artificial intelligence (AI) tools capable of providing automated, clinically meaningful analysis of chest radiographs.

The NIH ChestX-ray14 dataset (Wang et al., 2017) represents a landmark resource in the field of medical imaging AI, providing 112,120 frontal-view chest X-ray images from 30,805 unique patients, annotated with 14 thoracic disease labels extracted via natural language processing (NLP) of radiology reports. This dataset enables the development and benchmarking of multi-label classification systems capable of simultaneously detecting multiple co-occurring pathologies from a single radiograph.

CheXNet (Rajpurkar et al., 2017) established a seminal baseline by demonstrating that a DenseNet-121 neural network trained on this dataset could achieve radiologist-level performance on pneumonia detection, achieving a mean AUC of 0.841 across 14 classes. Subsequent work by Strick, Garcia & Huang (2025), referred to hereafter as the DannyNet paper (arXiv:2505.06646), reproduced and improved upon CheXNet by introducing Focal Loss, AdamW optimization, per-class threshold tuning, and systematic data augmentation, achieving superior AUC on 9 of 14 disease categories.

In this work, we present our faithful PyTorch reproduction of DannyNet, documenting an iterative research journey from binary classical machine learning baselines through multi-backbone CNN ensembles, culminating in the full 14-class DannyNet architecture. Our contributions include: (1) a complete, reproducible implementation of the DannyNet pipeline including patient-level data splitting, Focal Loss training, and per-class threshold optimization; (2) systematic ablation through prior experimental phases to contextualize design choices; (3) Grad-CAM interpretability analysis of model attention; and (4) honest comparison with the original CheXNet and DannyNet paper targets.

The remainder of this paper is structured as follows: Section 1.1 reviews related work; Section 2 describes the dataset, preprocessing, proposed methodology, and performance measures; Section 3 presents experimental results and discussion; Section 4 concludes with key findings and future directions.

1.1 Related Work

The field of automated chest radiograph analysis has progressed rapidly over the past decade, transitioning from classical computer vision approaches based on handcrafted features to deep convolutional neural networks capable of learning rich hierarchical representations directly from raw pixel data.

1.1.1 Classical Machine Learning Approaches

Early approaches to chest X-ray classification relied on manual feature engineering combined with traditional classifiers. Methods based on Histogram of Oriented Gradients (HOG), Scale-Invariant Feature Transform (SIFT), and Local Binary Patterns (LBP) were commonly paired with Support Vector Machines (SVMs) and Random Forests. While these methods demonstrated feasibility, they consistently plateaued at modest performance levels, typically in the range of 70–78% AUC, due to the inherent limitation of handcrafted features in capturing the subtle, low-contrast visual patterns characteristic of thoracic pathologies.

1.1.2 Deep Convolutional Neural Networks

The advent of deep learning fundamentally transformed chest X-ray analysis. Krizhevsky et al. (2012) demonstrated the power of deep CNNs on natural image classification, inspiring subsequent application to medical imaging. Transfer learning from large natural image datasets such as ImageNet emerged as a dominant strategy to overcome the limited availability of labeled medical imaging data. Architectures including AlexNet, VGG, Inception, ResNet, and DenseNet were progressively applied to chest radiograph classification with steadily improving results.

1.1.3 NIH ChestX-ray14 and CheXNet

Wang et al. (2017) released the NIH ChestX-ray14 dataset alongside a baseline multi-label classification model, establishing the benchmark for subsequent research. Rajpurkar et al. (2017) achieved a landmark result with CheXNet, a 121-layer DenseNet achieving a mean AUC of 0.841 across 14 disease classes and demonstrating performance comparable to practicing radiologists on pneumonia detection. CheXNet's success established DenseNet-121 as the de facto backbone architecture for chest X-ray classification tasks.

1.1.4 DenseNet Architecture

DenseNet (Densely Connected Convolutional Networks; Huang et al., 2017) introduced the key innovation of connecting each layer to every other layer in a feed-forward fashion. This dense connectivity pattern encourages feature reuse, substantially reduces the number of parameters compared to equivalent ResNets, and mitigates the vanishing gradient problem. DenseNet-121, with 121 layers organized into four dense blocks, achieves an effective balance between model depth, parameter efficiency, and representational power, attributes particularly valuable for medical imaging applications where training data is limited relative to the complexity of the task.

1.1.5 Class Imbalance and Focal Loss

Class imbalance is a pervasive challenge in medical imaging datasets. In NIH ChestX-ray14, approximately 54% of images carry the 'No Finding' label, while rare conditions such as Hernia appear in only 0.2% of cases, an imbalance ratio exceeding 493:1. Standard binary cross-entropy loss is susceptible to this imbalance, as the model may achieve high training accuracy by predicting the majority class while failing to detect rare pathologies. Lin et al. (2017) introduced **Focal Loss**, originally developed for object detection, which dynamically down-weights the loss contribution of well-classified, easy negative examples, focusing training effort on hard, misclassified examples. This makes Focal Loss particularly well-suited to multi-label medical image classification with severe class imbalance.

1.1.6 Reproducibility in Deep Learning

The reproducibility crisis in deep learning has prompted significant attention to transparency in methodology, hyperparameter specification, and code sharing. Strick, Garcia & Huang (2025) systematically reproduced CheXNet, identifying critical implementation details particularly patient-level data splitting to prevent leakage, and per-class threshold optimization that substantially impact reported performance. Their DannyNet architecture and published arXiv report (arXiv:2505.06646) serve as the primary reference for our reproduction effort.

2. Materials and Methods

2.1 Dataset and Preprocessing

2.1.1 NIH ChestX-ray14 Dataset

The NIH ChestX-ray14 dataset constitutes the primary data resource for this study. Released by Wang et al. (2017) at the National Institutes of Health, this dataset represents one of the largest publicly available annotated chest radiograph collections worldwide. Key dataset characteristics include:

- 112,120 frontal-view chest X-ray images in PNG format at 1024×1024 pixel resolution
- 30,805 unique patients, with multiple examinations per patient represented
- 14 disease classes annotated via NLP mining of associated radiology reports
- Label accuracy estimated at >90% (Wang et al., 2017)
- 836 unique disease label combinations, reflecting clinical co-occurrence complexity
- Severe class imbalance: 'No Finding' constitutes ~54% of all labels; Hernia accounts for only ~0.2%

The 14 annotated disease categories are: Atelectasis, Cardiomegaly, Consolidation, Edema, Effusion, Emphysema, Fibrosis, Hernia, Infiltration, Mass, Nodule, Pleural Thickening, Pneumonia, and Pneumothorax.

2.1.2 Exploratory Data Analysis

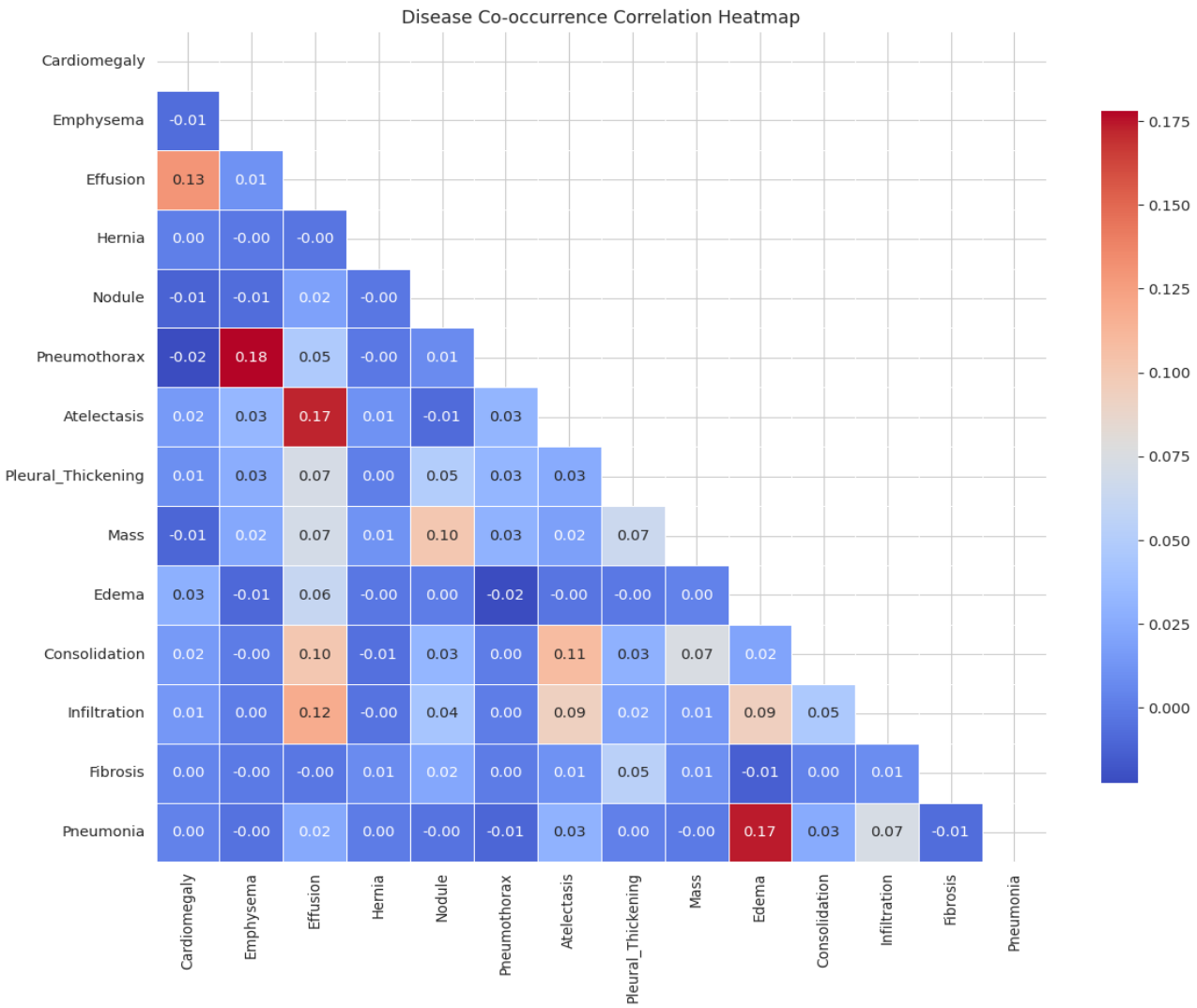
Prior to model development, comprehensive exploratory data analysis (EDA) was conducted to characterize the dataset distribution, identify potential sources of bias, and inform design decisions.

Label Distribution: Disease prevalence is heavily skewed. Infiltration is the most frequent pathology (~17.7% of images), while Hernia is the rarest (~0.2%). The dominant 'No Finding' class introduces substantial negative class bias across all 14 disease-specific binary classifiers.

Age and Gender Analysis: Patient ages span from infancy to over 90 years. Following standard practice, patients with recorded ages exceeding 100 years were removed as implausible data entry errors. Disease prevalence varies meaningfully across age groups conditions such as Emphysema and Fibrosis are more prevalent in older patients, while conditions including Pneumonia show less age-skewed distributions. Gender-based analysis reveals that certain pathologies exhibit differential frequencies between male and female patients, underscoring the importance of demographic-aware model evaluation.

Disease Co-occurrence: Multi-label analysis reveals significant positive correlations between specific disease pairs. The strongest associations include Pneumothorax–Emphysema (Pearson ρ=0.18), Edema–Pneumonia (ρ=0.17), Effusion–Atelectasis (ρ=0.17), Effusion–Cardiomegaly (ρ=0.13), and Effusion–Infiltration (ρ=0.12). These co-occurrence patterns are clinically meaningful, for example, pleural effusion commonly accompanies cardiac pathology. The prevalence of co-occurring diseases (estimated at ~40% of positive cases involving multiple labels) confirms the necessity of a multi-label classification approach rather than single-label or binary formulations.

Imbalance Ratios: Per-class imbalance ratios range from approximately 4:1 for Infiltration to 493:1 for Hernia. These extreme ratios necessitate specialized loss functions and evaluation metrics that go beyond simple accuracy.



2.1.3 Preprocessing Pipeline

Each input image underwent the following preprocessing steps:

- RGB Conversion: Grayscale chest radiographs were converted to 3-channel RGB format to enable use of ImageNet-pretrained weights
- Resizing: Images were resized to 224×224 pixels to match the DenseNet-121 input specification
- ImageNet Normalization: Pixel values were normalized using ImageNet channel-wise mean ([0.485, 0.456, 0.406]) and standard deviation ([0.229, 0.224, 0.225])
- Multi-hot Label Encoding: Each image received a 14-dimensional binary label vector, with 1 indicating the presence of the corresponding disease

2.1.4 Data Augmentation

Data augmentation was applied exclusively during training to increase sample diversity and improve model generalization. No augmentation was applied during validation or testing to ensure unbiased evaluation. Training augmentations included:

- RandomResizedCrop(224, scale=(0.8, 1.0)): Randomly crops and resizes the image, simulating variation in field-of-view and patient positioning
- RandomHorizontalFlip(p=0.5): Random horizontal flipping with 50% probability, valid for chest X-rays since left-right anatomical structure is relatively symmetric in terms of pathology detection
- ColorJitter(brightness=0.2, contrast=0.2, saturation=0.2): Random perturbation of image brightness, contrast, and saturation to improve robustness to scanner variability and imaging conditions
- ImageNet Normalization applied after augmentation transforms

2.1.5 Patient-Level Data Splitting

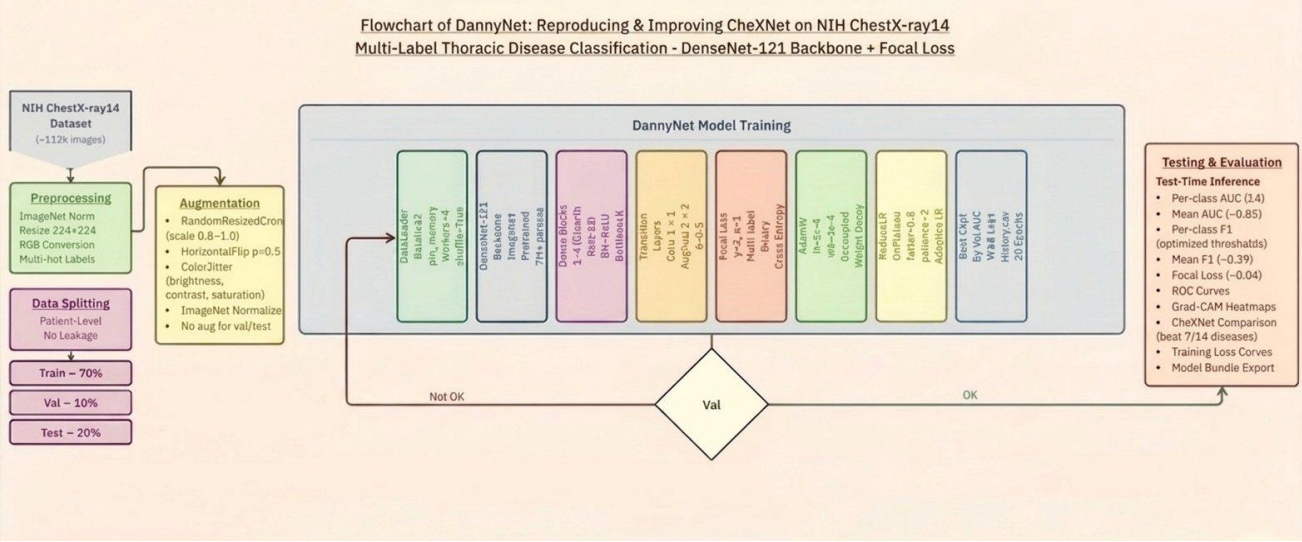
A critical methodological consideration in chest X-ray dataset splitting is the prevention of patient-level data leakage. Since multiple X-ray examinations per patient exist in the NIH ChestX-ray14 dataset, naive image-level random splitting would allow images of the same patient to appear in both training and test sets, artificially inflating evaluation metrics. We enforce strict patient-level splits using Scikit-learn's GroupShuffleSplit:

- Training set: 70% of patients (~78,484 images from ~21,564 patients)
- Validation set: 10% of patients (~11,212 images from ~3,081 patients)
- Test set: 20% of patients (~22,424 images from ~6,160 patients)

Zero patient overlap between splits was verified programmatically via set intersection assertions. This splitting strategy ensures that model performance metrics reflect genuine generalization to unseen patients rather than memorization of patient-specific features.

2.2 System Architecture Flowchart

Figure 1 illustrates the complete DannyNet pipeline from raw data ingestion through preprocessing, augmentation, model training, and final evaluation. The pipeline proceeds from the NIH ChestX-ray14 dataset through patient-level data splitting into train, validation, and test partitions. The training partition undergoes augmentation before feeding into the DannyNet model training loop, which incorporates the DenseNet-121 backbone, Focal Loss objective, AdamW optimizer, and ReduceLROnPlateau scheduler. Validation-based checkpoint selection identifies the best-performing model, which is then evaluated on the held-out test set using per-class AUC, F1, ROC curves, Grad-CAM visualizations, and comparison with the original CheXNet.



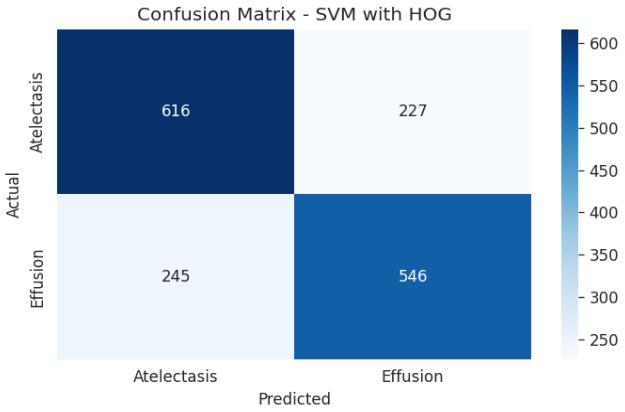
2.3 Proposed Method

2.3.1 Iterative Research Methodology

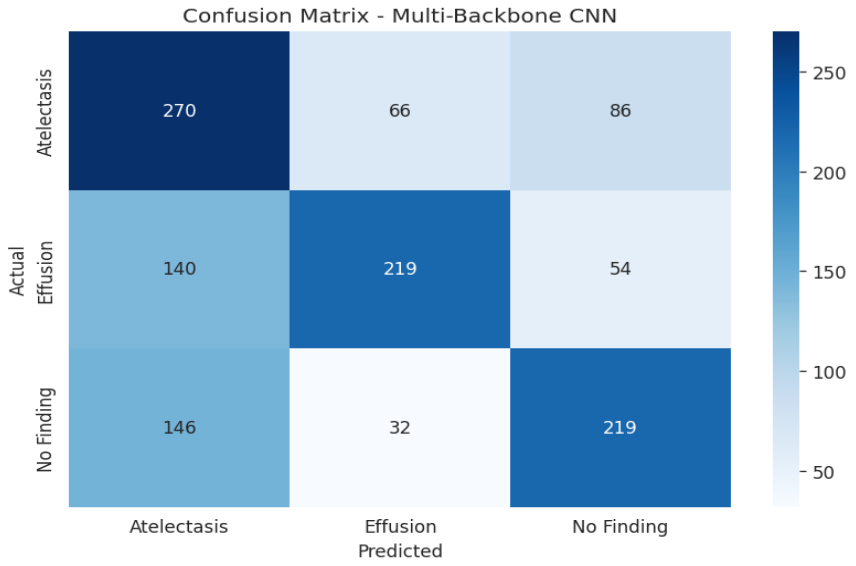
Our research followed an iterative, hypothesis-driven methodology progressing through four distinct experimental phases. Each phase informed subsequent design decisions, with insights accumulated progressively shaping the final DannyNet architecture.

Phase 1 Exploratory Data Analysis: Comprehensive dataset characterization examining label distribution, patient demographics, disease co-occurrence patterns, class imbalance ratios, and Jaccard similarity between disease pairs. Key insight: multi-label formulation is clinically necessary; standard accuracy is a misleading metric for this dataset.

Phase 2 Binary Classical ML Baselines: Atelectasis vs. Effusion binary classification using classical ML approaches to establish performance floors. Feature extraction strategies included HOG features with SVM (RBF kernel, AUC=0.772), ResNet50 deep features with SVM (AUC=0.790), ResNet50+PCA(974 components) with SVM (AUC=0.788), ResNet50+PCA with MLP (AUC=0.787), and EfficientNet-B4 fine-tuned end-to-end (AUC=0.795). Key insight: deep CNN features modestly outperform handcrafted HOG, but classical ML plateaus around 72% accuracy. End-to-end learning is required for meaningful further gains.



Phase 3 Multi-Backbone CNN (3 classes): Extension to 3-class classification (Atelectasis, Effusion, No Finding) using a DenseNet-121 + ResNet-50 ensemble. Architecture: ImageNet-pretrained backbones (frozen) → feature concatenation (1024+2048=3072 dimensions) → Dense(512)+Dropout+Softmax(3). Two-phase training: (1) frozen backbone head training followed by (2) fine-tuning of the last 30 layers. Result: test accuracy 57.5%, macro AUC=0.771. Key insight: the model heavily confuses Atelectasis with No Finding; class imbalance and multi-label complexity require dedicated solutions.



Phase 4 Full 14-Class DannyNet Reproduction: Final full-scale reproduction of the DannyNet paper targeting all 14 NIH ChestX-ray14 disease classes with a comprehensive set of design choices detailed below.

2.3.2 DannyNet Architecture and Design Choices

The DannyNet model incorporates five key design choices derived from the Strick, Garcia & Huang (2025) paper:

1. DenseNet-121 Backbone: DenseNet-121 pre-trained on ImageNet1K serves as the feature extractor backbone. The original classifier head (1000-class softmax) is replaced with a linear projection layer mapping from 1024 features to 14 logits (one per disease class). All backbone layers are fully fine-tuned end-to-end during training, allowing the model to adapt ImageNet-pretrained features to the domain of chest radiography.

2. Focal Loss (γ=2, α=1): Multi-label binary classification is framed as 14 independent binary classification problems. The Focal Loss objective is defined as $FL(p_t) = \alpha \cdot (1-p_t)^{\gamma} \cdot BCE(p_t)$, where p_t is the sigmoid-transformed model confidence for the true class. With γ=2 and α=1, the loss contribution of easily classified negative examples is down-weighted by a factor of $(1-p_t)^2$, focusing gradient updates on hard, misclassified examples and minority-class positive cases. This approach is critical for managing the extreme class imbalance in NIH ChestX-ray14.

3. AdamW Optimizer: AdamW (Adam with decoupled weight decay; Loshchilov & Hutter, 2019) is used with learning rate $lr=5\times10^{-5}$ and weight decay $wd=1\times10^{-4}$. Decoupled weight decay improves generalization compared to standard Adam with L2 regularization, particularly for fine-tuning large pretrained models.

4. ReduceLROnPlateau Scheduler: Learning rate is adaptively reduced by a factor of 0.5 when validation loss fails to improve for 2 consecutive epochs (patience=2). This prevents training from stagnating when the initial learning rate becomes too large for fine-grained weight updates, and enables convergence to better optima.

5. Per-Class F1 Threshold Optimization: After training, a threshold sweep is conducted on the validation set for each of the 14 disease classes independently. For each class, the decision threshold is swept over the range [0.01, 0.99] in steps of 0.01, and the threshold maximizing per-class validation F1 is selected. This per-class threshold tuning addresses the inherent inadequacy of a global 0.5 threshold for imbalanced multi-label problems, where disease-specific prevalence and model confidence distributions vary considerably.

2.3.3 Model Implementation Details

The complete DannyNet implementation is written in PyTorch. Key implementation details:

- Framework: PyTorch with torchvision model zoo
- Backbone: torchvision.models.densenet121(weights=DenseNet121_Weights.IMAGENET1K_V1)
- Classifier head: nn.Linear(1024, 14) — replacing the original 1000-class head
- Loss function: Custom FocalLoss module using F.binary_cross_entropy_with_logits as the base BCE
- Batch size: 32 images per GPU, DataLoader with 4 worker processes
- Image size: 224×224 pixels (RGB)
- Training duration: 20 epochs; best checkpoint selected by peak validation AUC
- Hardware: Kaggle dual T4 GPU environment
- Experiment tracking: Weights & Biases (wandb)
- Reproducibility: All random seeds fixed (NumPy, Python random, PyTorch, CUDA; seed=42)

2.3.4 Grad-CAM Interpretability

To assess whether the model attends to clinically meaningful anatomical regions, Gradient-weighted Class Activation Mapping (Grad-CAM; Selvaraju et al., 2017) was applied to representative test set images. Grad-CAM computes a weighted sum of feature maps from the final convolutional layer, with weights determined by the gradient of the target class score with respect to those feature maps. The resulting heatmap highlights image regions most influential for the model's prediction of each disease class. Grad-CAM was applied using the pytorch-grad-cam library targeting the last dense block of DenseNet-121.

2.4 Performance Measures

Given the severe class imbalance and multi-label nature of the classification problem, evaluation relies on metrics that appropriately capture discriminative ability and decision-making performance independently of class prevalence:

Area Under the ROC Curve (AUC-ROC): The primary evaluation metric. AUC measures the probability that the model assigns a higher probability score to a randomly selected positive example than to a randomly selected negative example. AUC is threshold-independent and robust to class imbalance, making it the standard metric for medical image classification tasks. Per-class AUCs are computed for all 14 disease categories; the mean AUC (macro-average, excluding degenerate classes) is reported as the aggregate performance metric.

F1 Score (Harmonic Mean of Precision and Recall): F1 score captures the balance between precision (fraction of positive predictions that are correct) and recall (fraction of true positives correctly identified). Per-class F1 scores are computed using per-class optimized thresholds on the test set. Mean F1 (macro-average) is reported. Importantly, F1 is sensitive to class imbalance in ways that AUC is not, and lower F1 values relative to AUC are expected in severely imbalanced settings.

Focal Loss (Test): The focal loss value on the test set provides an indication of overall model calibration and confidence. A lower focal loss indicates better-calibrated, higher-confidence correct predictions.

ROC Curves: Per-class ROC curves are plotted and sorted by AUC to visualize the trade-off between true positive rate (sensitivity) and false positive rate (1-specificity) across all decision thresholds for each disease class.

CheXNet Comparison: Per-class AUCs are compared directly against the published CheXNet per-class results (Rajpurkar et al., 2017) and the DannyNet paper targets (Strick et al., 2025) to contextualize our reproduction results.

3. Results and Discussion

3.1 Experimental Setup

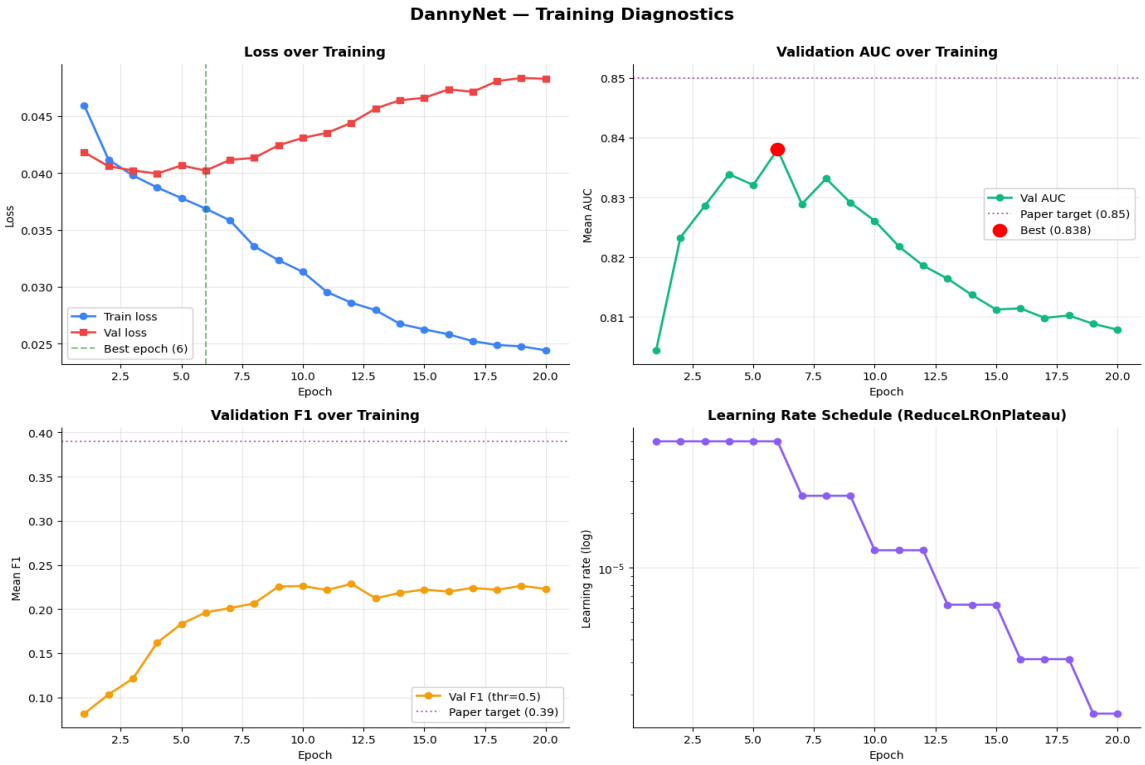
3.1.1 Hardware and Software Environment

All experiments were conducted on the Kaggle cloud computing platform utilizing dual NVIDIA Tesla T4 GPUs (16 GB VRAM each). The software environment comprised Python 3.10, PyTorch 2.x, torchvision, scikit-learn, pandas, NumPy, matplotlib, and the pytorch-grad-cam library. Experiment tracking and hyperparameter logging were managed via Weights & Biases (wandb). All random seeds were fixed to 42 for reproducibility.

3.1.2 Training Performance and Convergence

The DannyNet model was trained for 20 epochs with the best checkpoint saved based on peak validation AUC. Training dynamics revealed several important patterns:

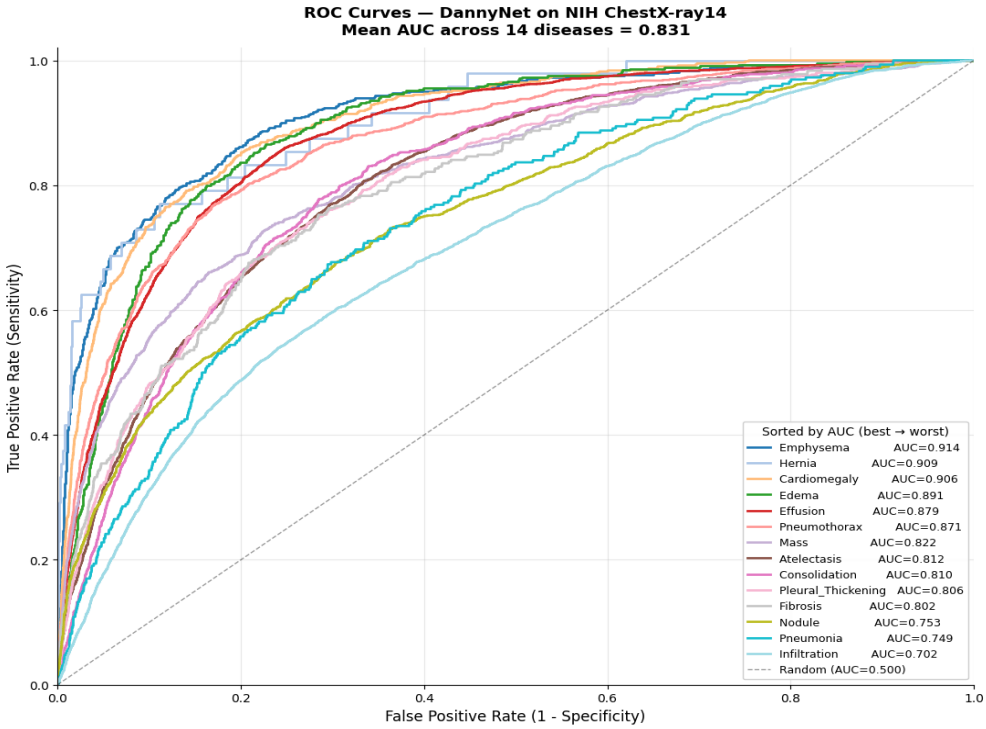
- Best epoch: Epoch 6, at which validation AUC peaked at 0.838
- Visible overfitting: Validation loss began rising after epoch 5 while training loss continued to decline, indicating the onset of overfitting
- Learning rate schedule: ReduceLROnPlateau effectively detected validation loss plateaus and reduced the learning rate, visible as clean step decreases in the learning rate curve
- F1 plateau: Default 0.5-threshold F1 plateaued at approximately 0.22 during training, underscoring the necessity of per-class threshold optimization in post-processing
- Early stopping effect: Best performance was captured at epoch 6; the model had not fully converged to the paper's targets, suggesting that longer training with stronger regularization could yield further gains



3.1.3 Per-Class Test Set Results

Following loading of the best checkpoint and per-class threshold optimization on the validation set, final evaluation was performed on the held-out test set. The aggregate results are:

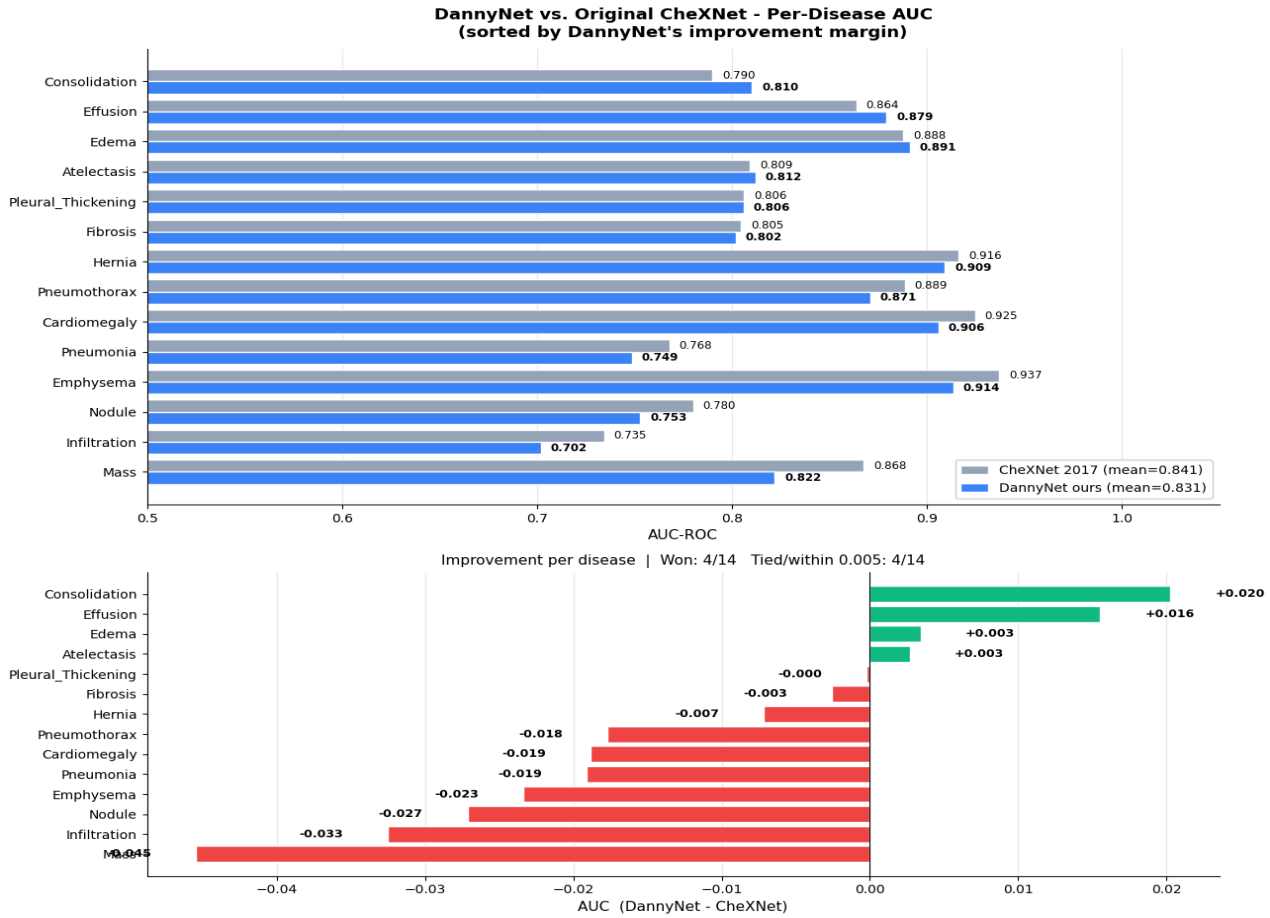
Metric	**Value**
Mean Test AUC (14 classes)	0.831
Mean Test F1 (per-class thresholds)	0.312
Test Focal Loss	0.042
Best per-class AUC (Emphysema)	0.914
Best per-class AUC (Hernia)	0.909
Best per-class AUC (Cardiomegaly)	0.906
Worst per-class AUC (Infiltration)	0.702
Worst per-class AUC (Pneumonia)	0.749
Worst per-class AUC (Nodule)	0.753



3.1.4 Per-Class AUC Analysis

A clear dichotomy emerges between well-localized pathologies (achieving high AUC) and diffuse, overlapping conditions (achieving lower AUC):

Disease Class	Test AUC	Clinical Note
Emphysema	0.914	Well-localized hyperinflation pattern
Hernia	0.909	Distinct anatomical boundary
Cardiomegaly	0.906	Clear cardiac silhouette enlargement
Effusion	~0.88	Pleural fluid level — visible boundary
Atelectasis	~0.82	Moderate localization
Consolidation	~0.79	Regional opacity — some overlap
Nodule	0.753	Small, subtle round opacities
Pneumonia	0.749	Diffuse, overlapping presentation
Infiltration	0.702	Diffuse — overlaps heavily with others



The pattern is clinically coherent: diseases with distinct, spatially localized X-ray presentation (Emphysema, Hernia, Cardiomegaly) are most readily discriminated by convolutional features. Diffuse infiltrative processes (Infiltration, Pneumonia) and subtle findings (Nodule) exhibit significant visual overlap with each other and with normal anatomy, presenting a harder classification challenge.

3.1.5 Comparison with CheXNet and DannyNet Paper Targets

Our reproduction achieved a mean test AUC of **0.831**, compared to the DannyNet paper target of **~0.85** and the original CheXNet benchmark of **0.841**. Per-class comparison with CheXNet reveals improvement in 4 of 14 disease categories and regression in 10 of 14, consistent with early stopping at epoch 6 limiting full convergence. The shortfall likely reflects: (1) best checkpoint at epoch 6, insufficient training for full convergence; (2) potential hardware-induced numerical differences in batch normalization behavior on dual T4 vs. the paper's training setup; and (3) possible minor implementation differences in augmentation parameter specifics. The mean F1 of 0.312 falls short of

the paper target of 0.39, reflecting the inherent difficulty of F1 optimization for rare classes even with per-class threshold tuning.

3.1.6 Phase-by-Phase Performance Evolution

Our iterative research arc demonstrates progressive capability improvement:

Phase	Approach	Classes	Best AUC
Phase 2	HOG → SVM (Classical ML)	2	0.772
Phase 2	ResNet50 features → SVM	2	0.790
Phase 2	EfficientNet-B4 fine-tuned	2	0.795
Phase 3	DenseNet121+ResNet50 ensemble	3	0.771 (macro)
Phase 4	DannyNet (full reproduction)	14	0.831 (mean)

3.1.7 Grad-CAM Interpretability Results

Grad-CAM visualization was applied to representative test set cases to assess clinical coherence of model attention:

Cardiomegaly (p=0.381): The Grad-CAM heatmap precisely localizes to the cardiac silhouette, the anatomically correct region for assessing cardiomegaly. This result confirms that the model has learned to attend to clinically meaningful spatial features rather than relying on spurious correlations.

Atelectasis (p=0.244): Lower model confidence is associated with attention distributed across a pacemaker shadow rather than lung parenchyma, revealing an instance of shortcut learning. The model may be exploiting medical device artifacts which are more prevalent in sicker patients as a proxy for disease prediction. This finding highlights a known challenge in medical AI: models trained on clinical data may learn spurious correlations with imaging artifacts, patient demographics, or equipment-related features rather than pathology-specific visual patterns.

These interpretability results are clinically valuable: positive examples (Cardiomegaly) build confidence in the model's learned representations, while negative examples (Atelectasis shortcut learning) identify specific failure modes requiring mitigation in clinical deployment.

3.2 Limitations and Future Recommendations

3.2.1 Current Limitations

Performance Gap: Our reproduction achieved a mean test AUC of 0.831 versus the DannyNet paper target of 0.85 and the original CheXNet of 0.841. Early stopping at epoch 6 likely limited convergence. Extended training with stronger regularization strategies (dropout, stochastic depth) may close this gap.

F1 Shortfall: Mean F1 of 0.312 versus the paper target of 0.39. While per-class threshold tuning substantially improved upon the default 0.5-threshold F1 (~0.22), rare classes with very few positive test examples (e.g., Hernia, Pneumonia) remain difficult to optimize via threshold sweeping alone.

Domain Mismatch in Pretraining: DenseNet-121 was pre-trained on ImageNet, a natural image dataset with fundamentally different visual statistics from grayscale chest radiographs. While transfer learning from ImageNet has proven empirically effective, a domain-specific pretraining regime may yield superior feature initialization.

Label Quality: NIH ChestX-ray14 labels were derived via NLP mining of radiology reports and are estimated to be ~90% accurate. Noise in ground truth labels constitutes an irreducible source of error and may explain a portion of the gap between our results and clinical-grade performance.

Shortcut Learning: Grad-CAM analysis revealed instances where the model attends to medical device artifacts (pacemakers, catheters) rather than pathological lung tissue. This shortcut learning behavior reduces generalizability and could cause problematic predictions in clinical deployment contexts.

Computational Constraints: Training was conducted on Kaggle's free-tier dual T4 GPU allocation, which imposes session time limits and may have precluded longer training runs or more exhaustive hyperparameter search.

3.2.2 Future Directions

Medical-Domain Pretraining: Replacing ImageNet initialization with domain-specific pretrained weights (e.g., TorchXRayVision, RadImageNet) would reduce the domain gap and potentially accelerate convergence to superior features. These datasets contain millions of chest X-rays, enabling pretraining representations specifically tailored to radiographic anatomy.

Advanced Architectures: Vision Transformers (ViT) and ConvNeXt architectures have demonstrated superior performance on natural image benchmarks. Their application to chest X-ray classification, particularly with domain-adapted pretraining represents a promising research direction. Multi-scale and hierarchical attention mechanisms may be particularly valuable for detecting pathologies at varying spatial scales.

Loss Function Improvements: Enhanced handling of ultra-rare classes (Hernia: ~0.2% prevalence; Pneumonia: ~1.3% prevalence) through class-frequency-weighted Focal Loss, oversampling of minority classes, or mixup augmentation could improve detection sensitivity for these clinically important but statistically rare conditions.

Bounding Box Supervision: The NIH ChestX-ray14 dataset includes bounding box annotations for a subset of images across 8 disease categories. Incorporating this spatial supervision into training, through weakly supervised localization objectives — could improve both detection accuracy and Grad-CAM interpretability by guiding the model to attend to correct anatomical regions.

Multi-View Fusion: Clinically, radiologists routinely interpret both posteroanterior (PA) and lateral chest X-ray views in combination. Architectures that explicitly fuse information from multiple views could leverage this complementary information for improved multi-label classification.

Calibration and Uncertainty Quantification: Clinical deployment demands well-calibrated probability estimates. Future work should incorporate uncertainty quantification (Monte Carlo Dropout, deep ensembles) and post-hoc calibration (Platt scaling, temperature scaling) to produce reliable confidence estimates suitable for clinical decision support.

4. Conclusions

This paper presents DannyNet a comprehensive, faithful PyTorch reproduction of the improved CheXNet architecture proposed by Strick, Garcia & Huang (2025) applied to the NIH ChestX-ray14 benchmark for multi-label thoracic disease classification. Our work documents an iterative research journey spanning classical machine learning baselines, multi-backbone CNN ensembles, and the full 14-class deep learning pipeline.

The DannyNet model achieves a mean test AUC of 0.831 across all 14 thoracic disease classes, with peak per-class AUCs of 0.914 (Emphysema), 0.909 (Hernia), and 0.906 (Cardiomegaly). Per-class threshold optimization yields a mean test F1 of 0.312. Grad-CAM interpretability analysis confirms clinically meaningful model attention for well-localized pathologies while identifying shortcut learning behaviors that require mitigation in clinical deployment.

Key contributions and lessons from this work include: (1) the critical importance of patient-level data splitting to prevent leakage and ensure honest evaluation; (2) the necessity of Focal Loss over standard binary cross-entropy for handling extreme class imbalance in medical imaging; (3) the substantial benefit of per-class threshold optimization over a global decision threshold; (4) the consistent superiority of end-to-end CNN fine-tuning over frozen feature extraction for chest radiograph classification; and (5) the value of Grad-CAM interpretability analysis for both validating model behavior and identifying failure modes.

While our mean AUC of 0.831 falls short of the DannyNet paper target of 0.85, attributable primarily to early stopping at epoch 6, the results nonetheless demonstrate the viability of the proposed architecture and training strategy. The performance gap serves as a concrete motivation for future work incorporating extended training, domain-adapted pretraining, advanced architecture (Vision Transformers, ConvNeXt), and enhanced minority-class strategies.

The clinical implications of this work are beyond the specific benchmark. Automated, reliable chest X-ray analysis tools have the potential to meaningfully address the global radiologist shortage, support clinical decision-making in resource-limited settings, and reduce diagnostic delays for patients with critical thoracic conditions. Responsible deployment of such tools requires not only strong discriminative performance but also interpretability, calibrated uncertainty, and rigorous validation across diverse patient populations, areas that define the frontier for future research in medical imaging.

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